Individual assignment 1 – IoT fundamentals

Section A – Multiple choice (20 marks – 10 questions x 2 marks)

1. Which statement best describes a complete IoT system?

   A. A sensor that measures temperature

   B. A microcontroller running a Python program

   C. A system that senses, communicates, processes data, and supports an application

   D. Any device connected to Wi-Fi

2. Which device primarily converts a physical quantity into measurable data?

   A. Actuator

   B. Sensor

   C. Router

   D. MQTT broker

3. A sensor records one temperature value every 30 seconds.

   A. network traffic

   B. time-series data

   C. an IP address

   D. an MQTT topic

4. Which is an example of a sensor-data quality problem?

   A. Missing measurement

   B. IPv6 address

   C. Wi-Fi connection

   D. MQTT subscription

5. What is the main purpose of serialization?

   A. To increase wireless range

   B. To convert structured data into a representation suitable for transmission or storage

   C. To select an IP address

   D. To increase sensor accuracy

6. A fitness sensor sends a small amount of data to a nearby smartphone and must operate from a small battery. Which connectivity technology is most appropriate?

   A. Ethernet

   B. BLE

   C. Satellite

   D. LoRa for every case

7. Which statement about IEEE 802.15.4 and Zigbee is correct?

   A. IEEE 802.15.4 is built on Zigbee

   B. They are exactly the same technology

   C. Zigbee can use IEEE 802.15.4 as its low-power PHY/MAC foundation

   D. Zigbee is a transport protocol such as TCP

8. Which transport protocol provides a connection-oriented, reliable, ordered byte stream?

   A. IP

   B. UDP

   C. TCP

   D. MQTT

9. Which application protocol is based on publish–subscribe communication?

   A. HTTP

   B. MQTT

   C. IPv6

   D. Ethernet

10. An MQTT alarm message is important and should be delivered, but receiving the same alarm twice is acceptable.

    A. QoS 0

    B. QoS 1

    C. QoS 2 is always required

    D. MQTT has no delivery levels

Section B – Short Answer (30 marks – 5 questions x 6 marks)

11. IoT Device vs IoT System.

A smartwatch measures a user's heart rate and displays the measurement on its own screen.

Explain the difference between:

1. an IoT device, and
2. a complete IoT system.

Use the smartwatch example in your explanation

- *IoT device = physical/embedded component → sense, process, and communicate.*
- *Complete IoT system includes multiple components such as device, communication, data processing, and application.*
- *A smartwatch measuring locally → smart/connected device, but local measurement alone does not necessarily constitute the complete IoT system.*
- *If measurements are communicated to a remote service/application for processing or monitoring, it forms part of a wider IoT system.*

12. Sensor data

A room-temperature sensor produces:

10:00	22.5 °C
10:01	22.7 °C
10:02	missing
10:03	22.8 °C
10:04	22.9 °C

Answer:

a. What type of data is this?

Time-series data.

b. Identify one data-quality problem.

Missing reading at 10:02.

c. Explain one possible way an application could handle the problem.

Acceptable examples:

- mark it as missing;

- interpolate if appropriate;

- use previous/next measurement carefully;

- request a new measurement if possible;

- exclude it from analysis.

13. Data Representation

```
sensor_data = {
    "device": "room01",
    "temperature": 22.8
}
```

And:

```
{"device": "room01", "temperature": 22.8}
```

Explain:

a. What is the first representation?

Python dictionary / Python data structure

b. What is the second representation?

JSON representation

c. Why might serialization be required before transmitting the information?

Serialization converts program data into a representation that can be transmitted or stored and reconstructed/interpreted by another application.

14. TCP and UDP

Explain two differences between TCP and UDP.

Your answer should refer to:

- connection establishment;

- delivery behavior.

Do not simply state that one is "faster."

TCP

- connection-oriented;

- establishes a connection before exchanging data;

- reliable;

- ordered;

- retransmission/acknowledgment mechanisms;

- byte-stream transport.

UDP

- connectionless;

- datagrams can be sent without prior connection;

- no built-in reliability;

- no built-in retransmission;

- no guarantee of order.

15. Request–Response vs Publish–Subscribe

Explain the difference between:

1. request–response; and

2. publish–subscribe.

Give one suitable IoT example for each.

Request–response

Client requests a resource/action and server returns a response.

Examples:

- HTTP API;

- browser requesting machine status;

- CoAP sensor resource request.

Publish–subscribe

Publisher sends information to a topic/broker; subscribers receive matching messages.

Example:

- continuous MQTT sensor telemetry.

Section C – Applied Scenario Questions (40 marks – 2 questions x 20 marks)

16. Smart Greenhouse IoT System

A greenhouse contains battery-powered soil-moisture sensors distributed across a large farm.

Each sensor:

- sends a small measurement every 15 minutes;

- may be located several kilometres from the farm office;

- must operate for a long time using battery power;

- does not need to send images or video.

a. Identify three communication requirements that are important when selecting connectivity for these sensors. **(3 marks)**

Possible requirements:

- long range;

- low power consumption;

- small data volume;

- low data rate;

- infrequent transmission;

- infrastructure availability.

Any three appropriate answers.

b. Choose a suitable connectivity technology.

Possible examples include:

- LoRa

Give two reasons for your choice. **(5 marks)**

Reasons:

- kilometre-scale coverage;

- suitable for small, infrequent sensor messages;

- designed for low-power applications.

The objective is reasoned selection, not memorizing one compulsory answer.

c. The gateway receives:

```
{
   "sensor": "soil07",
   "moisture": 41.2
}
```

What is JSON providing in this system? **(3 marks)**

JSON provides: structured representation/serialization of the application data.

d. Suppose many greenhouse sensors continuously report measurements to:

- a dashboard;
- a database;
- an irrigation-control application.

Would request–response or publish–subscribe be more suitable?

Explain. **(4 marks)**

Publish–subscribe.

Reasons include:

- continuous sensor reporting;

- several consumers require the same information;

- producer and consumers can be decoupled.

e. Name a suitable application protocol for the communication model selected in part (d).

Explain the role of the broker if your selected protocol uses one. **(5 marks)**

MQTT Broker:

- receives publications;

- organizes them by topic;

- distributes matching messages to subscribers;

- decouples publishers from subscribers.

17. Campus Environmental Sensor

A university installs battery-powered environmental sensors inside buildings.

Each sensor:

- has limited memory and processing capability;
- sends small temperature and humidity messages;
- uses IEEE 802.15.4;
- should participate in an IPv6-based IoT network;
- is occasionally queried by a local application.

a. Which application protocol is suitable for a constrained request–response device? **(2 marks)**

CoAP

b. Which transport protocol is commonly used with this application protocol? **(2 marks)**

UDP

c. What technology can adapt IPv6 communication to an IEEE 802.15.4 constrained network? **(3 marks)**

6LoWPAN

d. Complete the protocol stack: **(5 marks)**

Environmental Application

↓

CoAP

↓

UDP

↓

IPv6

↓

6LoWPAN

↓

IEEE 802.15.4

e. Explain the role of each of the following: **(8 marks)**

- application protocol;
- transport protocol;

CoAP

Defines lightweight application-level request–response communication.

UDP

Provides transport using ports without TCP-style connection/reliability mechanisms.

Section D — Integrated IoT Communication (10 marks)

18 — Trace the Data

Consider the following IoT system:

Temperature Sensor

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Python application

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JSON

↓

Wi-Fi

↓

IP

↓

TCP

↓

MQTT

↓

Broker

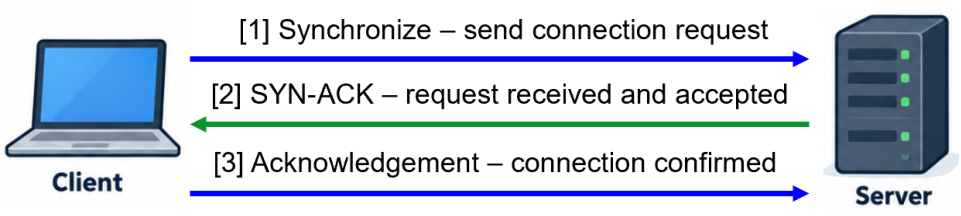
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Monitoring Application

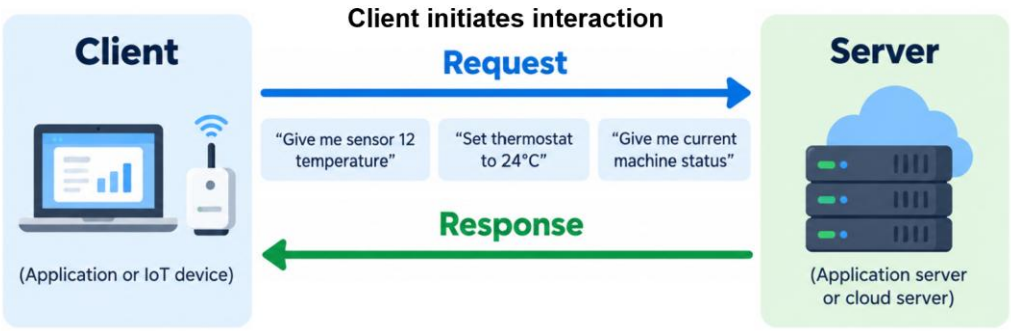
Explain the role of each of the following:

Technology	Expected role
JSON	Represents/serializes structured application data
Wi-Fi	Provides local wireless connectivity
IP	Provides network addressing and delivery between hosts/networks
TCP	Provides reliable ordered process-to-process byte-stream transport
MQTT	Provides application-level publish–subscribe messaging using topics and a broker

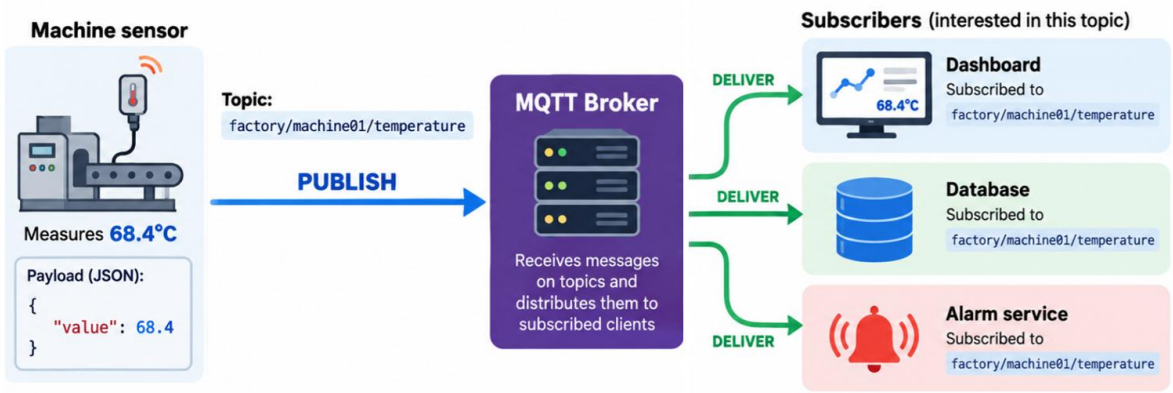
TCP handshakes:



Request-response:



Publish-subscribe:



- Sensor knows:
  - ✓ Broker address
  - ✓ Topic
  - ✓ Payload
- Sensor does not necessarily knows:
  - ✓ Dashboard address
  - ✓ Database address
  - ✓ Alarm-system address

Benefits:

- Sensor sends data to broker
- Broker distributes it to interested applications

➔ **MQTT QoS 0, 1, 2**

QoS selection:

QoS	Delivery behavior	Best for	Example	Trade-off
0	At most once	Frequent, non-critical data	Temperature telemetry	Lowest overhead, possible loss
1	At least once	Important messages	Fire alarm Machine fault alarm	More reliable, duplicates possible
2	Exactly once*	Messages where duplicate MQTT delivery should be avoided	Critical control message Financial transaction	Highest overhead

Question	HTTP	CoAP	MQTT
Main model	Request–response	Request–response	Publish–subscribe
Common transport	TCP	UDP	TCP
Architecture	Client–server	Client–server/resource	Broker based
Strength	Web/API integration	Constrained resources	Telemetry/event distribution
Typical IoT example	Cloud REST API	Constrained sensor	Continuous sensor messaging

Selection of connectivity technologies:

Technology	Range	Data	Power	Mobility	Dependency
Ethernet	Cable	Very high	powered	none	Cabling
Wi-Fi	Local	High	Medium–high	Moderate	AP
BLE	Short	Low–medium	Very low	Good	Phone/ gateway
802.15.4	Short–medium	Low	Low	Limited	Coordinator/ gateway
LoRa	Long	Very low	Very low	Limited/moderate	Gateway
NB-IoT	Wide	Low	Low	Limited	Cellular operator
LTE-M	Wide	Low–medium	Low–medium	Good	Cellular operator
4G/5G	Wide	High	Higher	Excellent	Cellular operator
Satellite	Very wide	Low–medium	Higher	Good	Satellite service